

Nikhil Vidhani

Lead Data Scientist – Fulfilment Simulation Strategy, Platforms & AI Enablement

Executive Summary

- Set the strategy and portfolio roadmap for Target's end-to-end fulfilment simulation capabilities, spanning platform architecture, accuracy, user experience, algorithm research, cloud-migration planning, and AI-enabled interfaces, with decisions potentially affecting up to \$8B in digital sales.
- Lead a cross-functional consolidation-revamp programme across product, engineering, data science, and fulfilment, now in A/B-test preparation with projected benefits of \$20M+ in shipping-cost savings and \$300M+ in incremental sales.
- Scaled Nemo, Target's end-to-end simulation platform for order allocation and available-to-promise, into the primary scenario-testing environment for more than 100 stakeholders, owning the release roadmap and operating mechanisms that improved accuracy, runtime, reliability, and self-service adoption.
- Lead AI enablement as AI Champion for Target's Digital Fulfilment data-science organisation, taking Nemo AI, a production multi-agent research and diagnosis system, from concept to internal production and establishing reusable capabilities and standards for adoption, evaluation, observability, governance, and human accountability.
- Develop technical capability and distributed ownership across a five-person Target team through onboarding, delegation, coaching, reviews, and knowledge transfer; previously led a five-member data-science team at WNS.

Industry and Research Experience

Nov 2022 – **Lead Data Scientist**, *Target Corporation, Bangalore*, Team Size: 5.

Present

- Own the product and technical strategy for simulation, optimisation, and AI enablement within Digital Fulfilment, translating business goals into sequenced roadmaps across platform capability, accuracy, reliability, user experience, algorithm research, and adoption.
- Lead complex cross-functional programmes from opportunity framing and business-case development through solution design, engineering alignment, validation, release planning, and controlled launch, balancing customer speed, cost, capacity, and sales outcomes.
- Establish operating mechanisms for roadmap governance, release cadence, quality measurement, self-service support, and recurring stakeholder communication to reduce one-off firefighting and shift capacity toward research and value creation.
- Serve as *AI Champion* for Target's Digital Fulfilment data-science organisation—setting the AI enablement agenda, leading production use cases, developing reusable Model Context Protocol (MCP) tools and AI skills, and creating forums and standards for responsible, measurable adoption.
- Provide technical and delivery leadership to a five-person team, onboarding and coaching scientists, delegating progressively larger workstreams, reviewing outcomes, and aligning individual ownership with platform and programme priorities; partner closely with product, engineering, supply-chain, and fulfilment leaders.

- Feb 2022 – **Sr. Group Manager (Data Science)**, *WNS Global Services, Bangalore*, Team Size: 5.
- Nov 2022
- Led a five-member data-science team delivering analytics and machine-learning solutions for finance and accounting use cases (payments, collections, reporting, prediction, and auditing) for global clients, driving multi-million-dollar operational savings.
 - Owned client and delivery outcomes from problem framing and solution prioritisation through data engineering, modelling, API development, and automated deployment, ensuring reliable and maintainable products.
 - Established team operating practices across shared repositories, code quality, CI/CD, reviews, technical sessions, and structured coaching, improving consistency and developing junior talent.
- Jun 2016 – **Research Scholar**, *Indian Institute of Management, Bangalore*, IC.
- Jan 2022
- Researched asset-pricing anomalies and examined the impact of disagreement on trading volume, analyst forecasts, and 10-K document characteristics.
 - Conducted comprehensive data collection, cleaning, modelling, and statistical analysis.
 - Published multiple papers and presented findings at several conferences. *Complete list available at: publications_CV.pdf*
- Jun 2018 – **Primary Instructor**, *Programming and Data Analysis*, Batch Size: 5–40.
- Jan 2022
- Designed and taught multiple short- and long-format courses in R programming, data analysis, and LaTeX.
 - Instructor rating: 4.7/5 (<https://github.com/nik141088/applied-R>).
- Jul 2012 – **Software Engineer**, *Cisco Systems, Bangalore*, IC.
- May 2016
- Designed, implemented, tested, reviewed, and documented software for 4G-LTE and Wi-Fi systems.
 - Contributed to high-availability, redundancy, and network-security architecture.

Major Projects

- 2026 – **Nemo AI & Generative AI Enablement**, *Target*.
- Present
- Set the vision and led **Nemo AI** from exploratory concept to an internally deployed multi-agent product for simulation research and diagnosis, prioritising high-friction support and analysis workflows with clear user and productivity goals.
 - Established production readiness across multi-environment authentication, cost and latency transparency, persisted history, full traces, and evaluation planning, creating the controls needed for reliable and governed adoption.
 - Built reusable organisational capabilities—a citation-first “LLM Wiki”, an MCP service over core systems, and simulation root-cause-analysis skills—so AI enablement could extend beyond one chatbot and remain traceable to source information.
 - Created a weekly AI-adoption cadence for the Digital Fulfilment data-science team, sharing wins and failures, removing blockers, and setting expectations for specifications, testing, prompt versioning, observability, evaluation, and human accountability.
 - Used the programme to develop team capability, onboarding a scientist into output analysis and promoting reusable skills, code-review workflows, and practical AI-assisted development across team workstreams.

2025 – Present	Consolidation Algorithm Revamp, Target. ○ Led the cross-functional revamp of Target’s legacy consolidation algorithm, aligning product, engineering, data science, and fulfilment on a common vision to improve customer promise speed, shipping cost, conversion, and margin. ○ Built and secured alignment on the business case, translating economic trade-offs into measurable speed, cost, sales, and No-Rush outcomes and quantifying projected benefits of \$20M+ in shipping-cost savings and \$300M+ in incremental digital sales. ○ Defined a staged validation and launch plan—algorithmic parity, end-to-end functional validation, mirror testing, and controlled A/B testing—raising data-science/engineering agreement from 20% to 95% and selecting 10 launch ZIP codes. ○ Set priorities across multiple rollout modes, No-Rush and network-priority levers, sequencing quick evidence-generating tests before larger integrations and managing unresolved product and engineering risks transparently. ○ Developed capability across the team and partner organisation by transferring algorithm internals to engineering and delegating test-case and coordination ownership to a scientist while maintaining review and quality standards. ○ Maintained hands-on technical oversight of objective design, simulation evidence, implementation defects, and latency/reliability constraints, enabling phased rollout decisions through decision documents, deep dives, and knowledge-transfer sessions.
2023 – Present	Nemo: Unified Simulation Platform, Target. ○ Set the vision and roadmap for Nemo as a unified simulation platform for last-mile promise and fulfilment, supporting more than 100 stakeholders across planning, digital fulfilment, and operations. ○ Evolved the platform from a specialist tool into a managed product with shared data and date-computation engines across available-to-promise (ATP) and guest order allocation (GOA), user interfaces, CLI and automation paths, replay and checkpoint recovery, and a roadmap toward cloud and agentic interfaces. ○ Owned release and quality governance across five major and 12 minor releases in 2025–2026, introducing automated accuracy measurement, validation pipelines, self-service documentation, and run-status and reporting mechanisms. ○ Led platform-quality improvements that raised ATP accuracy from about 40% to 85%, increased before/after-cutoff agreement from 39% to 88% by resolving a systemic time-zone defect, and reduced runtime by 40%, while building repeatable mechanisms rather than one-off fixes. ○ Led simulation studies informing decisions affecting roughly 3% of Target’s digital orders (\$600M+), including an evaluation identifying up to \$5M in potential shipping-cost savings for an estimated 0.03-day increase in delivery time. ○ Developed team ownership across accuracy, UI/backend, releases, user support, and migration workstreams; drove adoption through user enablement, a 30-person knowledge-transfer session, leadership deep dives, and an organisation-wide Simulations Conference.
2021 – 2024	Past Projects, Target, WNS, IIMB, and Personal. <i>A list of selected past projects: selected_past_projects_CV.pdf</i>

Tools and Skills

Expert	Programming & Data Engineering , Python, R, PySpark, SQL, Kafka, Hive/Hadoop, distributed data pipelines, data modelling, Shiny/Streamlit, C/C++.
Advanced	Analytics & Modelling , Simulation, operations research, optimisation, experiment design, A/B testing, regression and classification, NLP, scenario analysis, business-impact measurement.
Advanced	Generative AI & Agentic AI , AI product strategy, multi-agent orchestration, Model Context Protocol (MCP), LLM application design, evaluation, governance, observability, traceability, specification-driven development, adoption enablement.
Expert	Design Principles , Scalable system and algorithm design, reliability, performance, test-driven and modular development, secure coding, quality controls, and production-readiness standards.

Advanced	Engineering & Platforms , Platform and technical roadmaps, APIs and services (Plumber/FastAPI), distributed processing, authentication, observability, Linux, Git/GitHub, relational databases, AWS, cloud-migration planning, CI/CD.
Advanced	Visualisation & Storytelling , Executive and technical communication, decision documents, roadmap presentations, deep dives, ggplot2, Plotly, R Markdown, dashboarding, LaTeX, Confluence/Jira.
Expert	Domain Knowledge , Last-mile operations research (available-to-promise and order allocation), digital fulfilment, promise-speed and shipping-cost optimisation, finance, accounting, and statistics.
Advanced	Product & Strategy , Strategy and portfolio prioritisation, product thinking, roadmap design, business cases, ambiguous problem framing, impact measurement, stakeholder alignment, and controlled launch planning.
Advanced	Leadership , Technical team leadership, capability development, delegation and distributed ownership, coaching, cross-functional programme leadership, operating cadence, stakeholder influence, candidate interviewing, and knowledge transfer.

Education

2016 – 2021	Doctor of Philosophy , Finance and Accounting, Indian Institute of Management, Bangalore. CGPA: 3.64 / 4.00
2010 – 2012	Master of Engineering , Electrical Communication Engineering, Indian Institute of Science, Bangalore. CGPA: 6.4 / 8.0; Project Grade: A
2006 – 2010	Bachelor of Technology , Electronics and Communication Engineering, Bundelkhand Institute of Engineering and Technology, Jhansi (U.P.). Grade: 73.6%
2004 – 2006	Schooling (12th Standard) , Science, C.B.S.E.. Grade: 88.2%

Awards and Honours

2025	Drive , Target Corporation.
2022	Tech Genius , Transforming the Organization, Victory Fleet, WNS Global Services.
2020	Mirae Asset Scholarship (PhD Year 5) , Indian Institute of Management.
2017–2018	Director's Merit List (PhD Years 1 and 2) , Indian Institute of Management.
2016	96.5th percentile , Common Aptitude Test (CAT).
2010	All India Rank 20 of 105,000 , Graduate Aptitude Test in Engineering (GATE).
2006–2007	Merit Scholarships , Intermediate Examination and Engineering.